



[4]

SARDAR PATEL UNIVERSITY
BCA (SEMESTER-IV) EXAMINATION ,April-2022
US04FBCA01 :Computer Based Numerical & Statistical Methods (NC)
29/8/2022, Time :10:00 a.m. to 12:00 p.m.

Total Marks : 70
[10]

Q.1 Multiple Choice Questions.

- 1 For real root of an equation $x^3 - 2x - 5 = 0$, the root lies between
 (a) 0 and 1 (b) 2 and 3 (c) 1 and 2 (d) none of them
- 2 From the following _____ method is the best method to obtain root of equation $f(x) = 0$.
 (a) False position (b) Bisection (c) Newton's Raphson (d) none of them
- 3 All the formulae of interpolation are based on the fundamental assumption that the given data can be expressed as a _____.
 (a) Polynomial (b) Equation (c) Algorithm (d) None of the above
- 4 _____ method of interpolation is used for unequally spaced functions.
 (a) Forward difference (b) backward difference
 (c) Newton's divided difference (d) None of the above
- 5 _____ is not a type of interpolation method.
 (a) Forward difference (b) backward difference
 (c) Newton's divided difference (d) moving average method
- 6 The system of linear equation $AX = B$ can be solved by matrix inversion method only if...
 (a) $A \neq 0$ (b) $|A| \neq 0$ (c) $|A| = 0$ (d) A is symmetric
- 7 The system of linear equation $AX = B$ can be solved by Gauss-Seidel method only if...
 (a) All diagonal elements of A are zero
 (b) All diagonal elements of A are non zero
 (c) All diagonal elements of A are dominant
 (d) A is skew symmetric
- 8 One purpose of short-range forecasts is to determine _____.
 (a) production planning (b) inventory budgets
 (c) facility location (d) job assignments
- 9 Irregular variations in a time series are caused by:
 (a) lockouts and strikes (b) epidemics
 (c) floods (d) all the above
- 10 Forecasts are usually classified by time horizon into three categories
 _____.
 (a) short-range, medium-range, and long-range
 (b) finance/accounting, marketing, and operations
 (c) strategic, tactical, and operational
 (d) exponential smoothing, regression, and time series

Q.2 Fill in the Blanks and True or False.

[08]

- 1 The number 0.01850×10^3 has _____ significant digits .
- 2 _____ is called the forward difference operator.
- 3 Rate of change of distance with respect to time reprints _____
- 4 A time series consists of _____ components
- 5 Absolute error is defined as Present Approximation – Previous Approximation (True /False)

- 6 If population census for the years 1931, 1941, 1951, 1961 and 1971 is given and if we want to estimate the population for the year 1935 then Forward difference method is used. (True /False)
- 7 The system of linear equation $AX = B$ is said to be homogeneous if $B = 0$ (True /False)
- 8 Gradual, long-term movement in time-series data is called trends . (True /False)

Q.3 Answer the Following.(Any 10)

[20]

- 1 Define Relative error and absolute error.
- 2 Describe the stopping rules to obtain approximate solution for given non-linear equations.
- 3 Define Interpolation.
- 4 Draw forward difference table.
- 5 If $y = 2x^3 - x^2 + 3x + 1$, calculate the value of y corresponding to $x = 0, 1, 2, 3, 4, 5$ and form the backward difference table..
- 6 If x lies in the upper half of the table and if $x_k < x < x_{k+1}$, then what is $\frac{dy(x)}{dx}$ and $\frac{d^2y(x)}{dx^2}$?
- 7 Solve the following system of equations.
 $10x + 3y = 5$
 $8x - 2y = 2$
- 8 If x lies in the lower half of the table and if $x_{k-1} < x < x_k$, then what is $\frac{dy(x)}{dx}$ and $\frac{d^2y(x)}{dx^2}$?
- 9 What is Time Series?
- 10 List the utilities of time series.
- 11 What do you mean by Cyclic Variation?
- 12 What do you mean by random or irregular Variation?

Q.4 Answer the Following.(Any 4)

[32]

- 1 Find the correct root up to 3 decimal places by False position method for $x^3 - 2x - 5 = 0$
- 2 Find the correct root up to 3 decimal places by Newton Raphson method $x^2 - 12 = 0$
- 3 Estimate the expectation of life at the age of 16 years by using the following data:

Age (In Years)	10	15	20	25	30	35
Expectation of life (In Year)	35.4	32.3	29.2	26.0	23.2	20.4

- 4 The following table gives the census population of a town for the years 1931 to 1971. Estimate the population for the year 1965 by using an appropriate interpolation formula.

Year	1931	1941	1951	1961	1971
Population	46	66	81	93	101

- 5 Solve the following system of equations using matrix inversion method.

$$2x_1 - 2x_2 + 5x_3 = 13$$

$$2x_1 + 3x_2 + 4x_3 = 20$$

$$3x_1 - x_2 + 3x_3 = 10$$

- 6 The distance (s) covered by a car in given time (t) is given in the following table

Time(Miutes)	10	12	14	16	18
Distance(km)	12	15	20	27	37

Determine the speed of the car at t = 13 minutes.

- 7 Obtain the trend from the time series given below by method of moving average of
[i] 3 years [ii] 5 years

Year	1961	1962	1963	1964	1965	1966	1967	1968	1969	1970
Y	500	540	550	530	520	560	600	640	620	640

- 8 Obtain seasonal indices using simple average method.

Year	Q-I	Q-II	Q-III	Q-IV
1974	30	81	62	119
1975	33	104	86	171
1976	42	153	99	221
1977	56	172	129	235
